

Research Notes on Adversarial examples

Adversarial examples

>> Section 1

Targeted: $\min_{x'} d(x', x) \text{ subject to } C(x') = c$

>> Section 2

$\min_{\delta} ||\delta||_p + c \cdot f(x + \delta), x + \delta \in [0, 1]^n$

>> Section 3

Examples Examples include attacks in spam filtering, where spam messages are obfuscated through the misspelling of "bad" words or the insertion of "good" words; attacks in computer security, such as obfuscating malware code within network packets or modifying the characteristics of a network flow to mislead intrusion detection; attacks in biometric recognition where fake biometric traits may be exploited to impersonate a legitimate user; or to compromise users' template galleries that adapt to updated traits over time. Researchers showed that by changing only one-pixel it was possible to fool deep learning algorithms. Others 3-D printed a toy turtle with a texture engineered to make Google's object detection AI classify it as a rifle regardless of the angle from which the turtle was viewed. Creating the turtle required only low-cost commercially available 3-D printing technology. A machine-tweaked image of a dog was shown to look like a cat to both computers and humans. A 2019 study reported that humans can guess how machines will classify adversarial images. Researchers discovered methods for perturbing the appearance of a stop sign such that an autonomous vehicle classified it as a merge or speed limit sign. A data poisoning filter called Nightshade was released in 2023 by researchers at the University of Chicago. It was created for use by visual artists to put on their artwork to corrupt the data set of text-to-image models, which usually scrape their data from the internet without the consent of the image creator. McAfee attacked Tesla's former Mobileye system, fooling it into driving 50 mph over the speed limit, simply by adding a two-inch strip of black tape to a speed limit sign. Adversarial patterns on glasses or clothing designed to deceive facial-recognition systems or license-plate readers, have led to a niche industry of "stealth streetwear". An adversarial attack on a neural network can allow an attacker to inject algorithms into the target system. Researchers can also create adversarial audio inputs to disguise commands to intelligent assistants in benign-seeming audio; a parallel literature explores human perception of such stimuli. Clustering algorithms are used in security applications. Malware and computer virus

analysis aims to identify malware families, and to generate specific detection signatures. In the context of malware detection, researchers have proposed methods for adversarial malware generation that automatically craft binaries to evade learning-based detectors while preserving malicious functionality. Optimization-based attacks such as GAMMA use genetic algorithms to inject benign content (for example, padding or new PE sections) into Windows executables, framing evasion as a constrained optimization problem that balances misclassification success with the size of the injected payload and showing transferability to commercial antivirus products. Complementary work uses generative adversarial networks (GANs) to learn feature-space perturbations that cause malware to be classified as benign; Mal-LSGAN, for instance, replaces the standard GAN loss with a least-squares objective and modified activation functions to improve training stability and produce adversarial malware examples that substantially reduce true positive rates across multiple detectors.

>> Section 4

Adversarial attacks and training in linear models There is a growing literature about adversarial attacks in linear models. Indeed, since the seminal work from Goodfellow et al. studying these models in linear models has been an important tool to understand how adversarial attacks affect machine learning models. The analysis of these models is simplified because the computation of adversarial attacks can be simplified in linear regression and classification problems. Moreover, adversarial training is convex in this case. Linear models allow for analytical analysis while still reproducing phenomena observed in state-of-the-art models. One prime example of that is how this model can be used to explain the trade-off between robustness and accuracy. Diverse work indeed provides analysis of adversarial attacks in linear models, including asymptotic analysis for classification and for linear regression. And, finite-sample analysis based on Rademacher complexity. A result from studying adversarial attacks in linear models is that it closely relates to regularization. Under certain conditions, it has been shown that

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Repeat this for $v_2, v_3, \dots$ until the desired level of error in the classifier is reached. It was discovered when the authors designed a simple baseline to compare with a previous black-box adversarial attack algorithm based on gaussian processes, and were surprised that the baseline worked even better.

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$S(x') > 0 \iff \begin{cases} \arg \max_c F(x') \neq C(x), & \text{(Untargeted)} \\ \arg \max_c F(x') = c^*, & \text{(Targeted)} \end{cases}$

>> Section 7

Adversarial examples An adversarial example refers to specially crafted input that is designed to look "normal" to humans but causes misclassification to a machine learning model. Often, a form of specially designed "noise" is used to elicit the misclassifications. Below are some current techniques for generating adversarial examples in the literature (by no means an exhaustive list).

>> Section 8

Challenges in applying machine learning to security Researchers have observed that the constraints under which machine-learning techniques function in the security domain are different from those of common benchmark domains. Security data may change over time, include mislabeled samples, or reflect adversarial behavior, which complicates evaluation and reproducibility.

>> Section 9

Class imbalance In realistic deployment environments, the proportion of malicious samples can be extremely low, ranging from 0.01% to 2% of total data. This unbalanced distribution causes models to develop a bias towards the majority class, achieving high accuracy but failing to identify malicious samples. Prior approaches to this problem have included both data-level solutions and sequence-specific models. Methods like n-gram and Long Short-Term Memory (LSTM) networks can model sequential data, but their performance has been shown to decline significantly when malware samples are realistically proportioned in the training set, demonstrating the limitations in realistic security contexts. To address this issue, one approach has been to adapt models from natural language processing, such as BERT. This method involves treating sequences of application activities as a form of "language" and fine-tuning a pre-trained BERT model on the specific task. A study applying this technique to Android activity sequences reported an F1 score of 0.919 on a dataset with only 0.5% malware samples. This result was a significant improvement over LSTM and n-gram models, demonstrating the potential of pre-trained models to handle class imbalance in malware detection.